ANALOG AND DIGITAL COMMUNICATION
EXPERIMENT NO:3
MATLAB-Based Modelling of VSB Modulator and Demodulator for Analog Television Communication Applications

NAME:**FIONA PRISCILLA MIT S.** REG NO:**1912260145**

PROGRAM:

```
clc;
clear;
close all;

%% VSB-SC MODULATOR AND DEMODULATOR

%% 1. Parameters
Fs = 500000;
T = 0.02;
t = 0:1/Fs:T-1/Fs;
N = length(t);

fc = 100000;      % Carrier frequency
Ac = 1;           % Carrier amplitude

fm1 = 2000;
fm2 = 5000;
fm3 = 8000;

%% 2. Message / TV Video Signal
m = 0.5*cos(2*pi*fm1*t) + ...
   0.3*cos(2*pi*fm2*t) + ...
   0.2*cos(2*pi*fm3*t);

m = m / max(abs(m));

%% 3. Carrier
carrier = Ac*cos(2*pi*fc*t);

%% 4. DSB-SC Modulation
dsb_sc = m .* carrier;

%% 5. Frequency Axis
f = (-N/2:N/2-1)*(Fs/N);

%% 6. DSB-SC Spectrum
DSB = fftshift(fft(dsb_sc));
```

```
%% 7. VSB Filter
Bm = fm3;
Bv = 2000;

H = zeros(size(f));

% Positive frequencies
H(f >= fc & f <= fc + Bm) = 1;
H(f >= fc-Bv & f < fc) = 0.5;

% Negative frequencies
H(f <= -fc & f >= -fc-Bm) = 1;
H(f < -fc & f >= -fc+Bv) = 0.5;

%% 8. VSB Signal
VSB_spectrum = DSB .* H;

vsb = real(ifft(ifftshift(VSB_spectrum)));

%% 9. Add Noise
SNR_dB = 30;

signal_power = mean(vsb.^2);
noise_power = signal_power / (10^(SNR_dB/10));

noise = sqrt(noise_power) * randn(size(vsb));

received_vsb = vsb + noise;

%% 10. Coherent Detection
local_carrier = 2*cos(2*pi*fc*t);

detector_output = received_vsb .* local_carrier;

%% 11. Low Pass Filter
cutoff = Bm;

[b,a] = butter(6, cutoff/(Fs/2));

recovered_signal = filtfilt(b,a,detector_output);

%% 12. Normalize
recovered_signal = recovered_signal / max(abs(recovered_signal));

%% 13. Performance Analysis

MSE = mean((m - recovered_signal).^2);
```

```
R = corrcoef(m,recovered_signal);
Correlation = R(1,2);

signal_power = mean(m.^2);
error_power = mean((m-recovered_signal).^2);

SNR = 10*log10(signal_power/error_power);

%% 14. Display Results

fprintf('\n=====================================\n');
fprintf('      VSB SYSTEM RESULTS\n');
fprintf('=====================================\n');

fprintf('Carrier Frequency      = %.2f kHz\n',fc/1000);
fprintf('Message Bandwidth      = %.2f kHz\n',Bm/1000);
fprintf('Vestigial Bandwidth    = %.2f kHz\n',Bv/1000);
fprintf('VSB Bandwidth          = %.2f kHz\n',(Bm+Bv)/1000);

fprintf('\nPerformance:\n');
fprintf('MSE                    = %.6f\n',MSE);
fprintf('Correlation Coefficient = %.4f\n',Correlation);
fprintf('SNR                    = %.2f dB\n',SNR);

fprintf('=====================================\n');

%% 15. Plot Original Signal

figure;
plot(t*1000,m,'LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Original TV Video Signal');
grid on;
xlim([0 2]);

%% 16. Plot DSB-SC Signal

figure;
plot(t*1000,dsb_sc);
xlabel('Time (ms)');
ylabel('Amplitude');
title('DSB-SC Signal');
grid on;
xlim([0 0.5]);

%% 17. DSB-SC Spectrum
```

```
figure;
plot(f/1000,abs(DSB)/N);
xlabel('Frequency (kHz)');
ylabel('Magnitude');
title('DSB-SC Spectrum');
grid on;
xlim([-120 120]);

%% 18. VSB Filter Response

figure;
plot(f/1000,H,'LineWidth',1.5);
xlabel('Frequency (kHz)');
ylabel('Gain');
title('VSB Filter Response');
grid on;
xlim([-120 120]);

%% 19. VSB Spectrum

figure;
plot(f/1000,abs(VSB_spectrum)/N);
xlabel('Frequency (kHz)');
ylabel('Magnitude');
title('VSB-SC Spectrum');
grid on;
xlim([-120 120]);

%% 20. Received VSB Signal

figure;
plot(t*1000,received_vsb);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Received VSB Signal');
grid on;
xlim([0 0.5]);

%% 21. Product Detector Output

figure;
plot(t*1000,detector_output);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Product Detector Output');
grid on;
xlim([0 2]);
```

```
%% 22. Recovered Signal

figure;
plot(t*1000,recovered_signal,'LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Recovered TV Signal');
grid on;
xlim([0 2]);

%% 23. Original vs Recovered

figure;

plot(t*1000,m,'LineWidth',1.2);
hold on;

plot(t*1000,recovered_signal,'--','LineWidth',1.2);

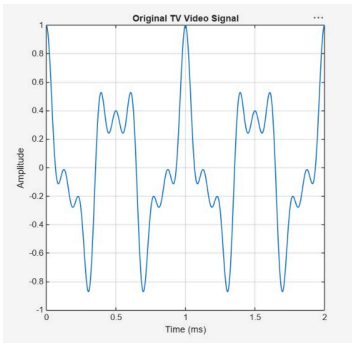
xlabel('Time (ms)');
ylabel('Amplitude');

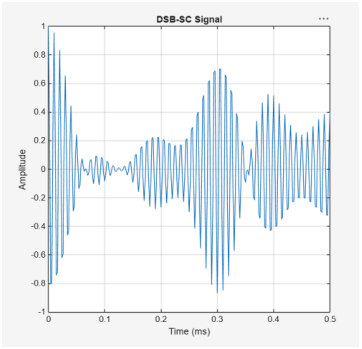
title('Original vs Recovered TV Signal');

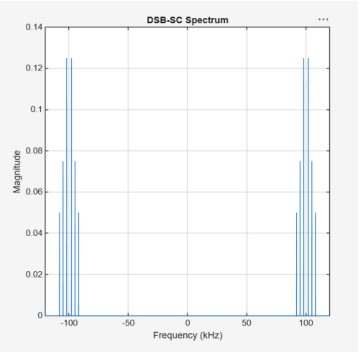
legend('Original Signal','Recovered Signal');

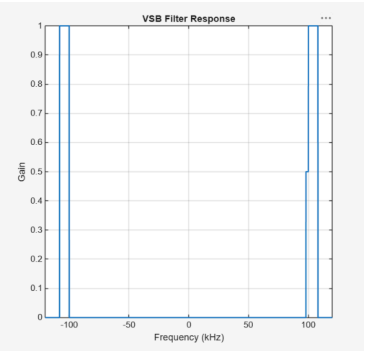
grid on;
xlim([0 2]);
```

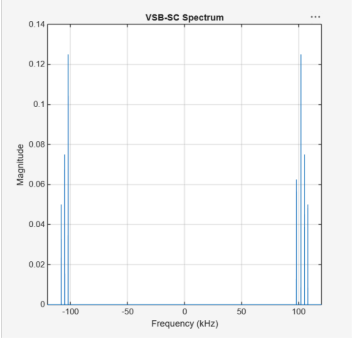
OUTPUTS:

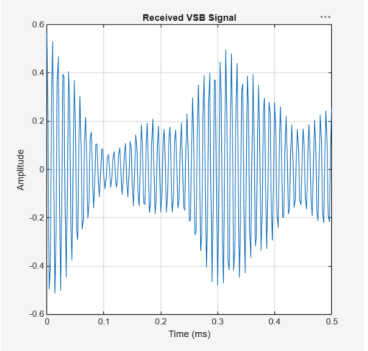


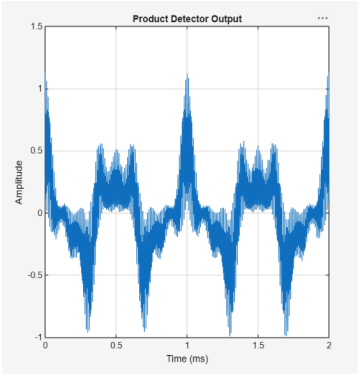


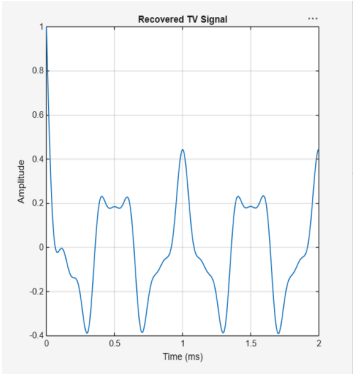


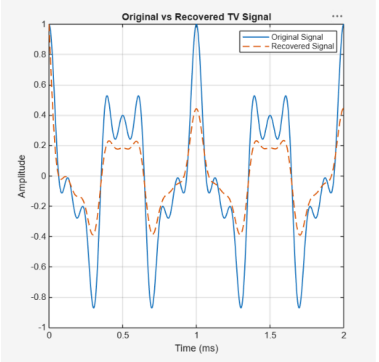












==========

VSB SYSTEM RESULTS

==========

Carrier Frequency = 100.00 kHz
Message Bandwidth = 8.00 kHz
Vestigial Bandwidth = 2.00 kHz
VSB Bandwidth = 10.00 kHz

Performance:
MSE = 0.052994
Correlation Coefficient = 0.9760
SNR = 5.55 dB

==========